Design Documentation

Interactive House – SG2: Units

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Revision History

Date	Version	Description	Author
25/Feb/26	1.0	Initial Design + Diagrams for Web (D1–D4)	Mustafa Al-Bayati
25/Feb/26	1.1	Initial Design + Diagrams for Mobile (D5–D7)	Daniel Jönsson
15/Mar/26	1.2	Updated D3 – D4 Add Mobile App Pictures (D5 – D6)	Mustafa Al-Bayati & Daniel Jönsson
21/Apr/26	1.3	Add Table of Contents, Refactor	Daniel Jönsson

Design Item List

Requirement Name	Priority
D1. Web Welcome Screen	Desirable
D2. Web User Authentication (Login/Signup)	Essential
D3. Web Device Management Interface	Essential
D4. Web Hub Control Panel	Essential
D5. Mobile Login/Signup	Essential
D6. Mobile Control Devices in Device Hub	Essential
D7. Mobile Database Connection	Essential

Design Item Descriptions

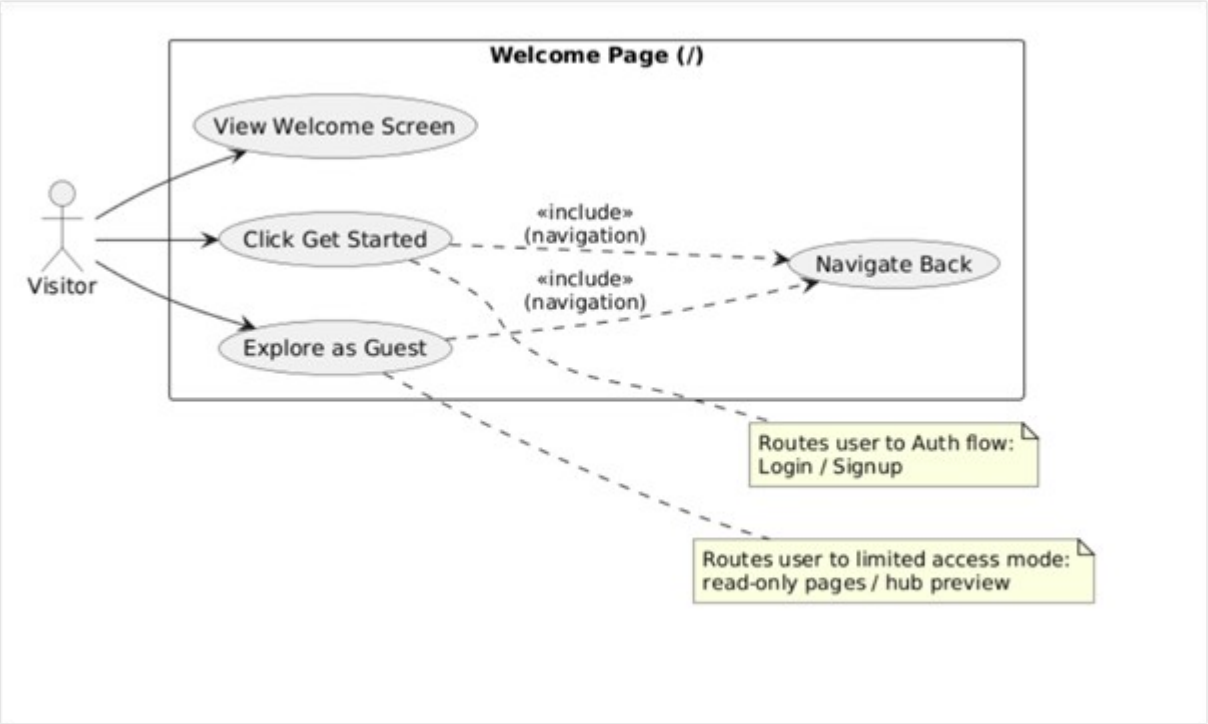
1. Web (Next.js + React)

D1. Web Welcome Screen

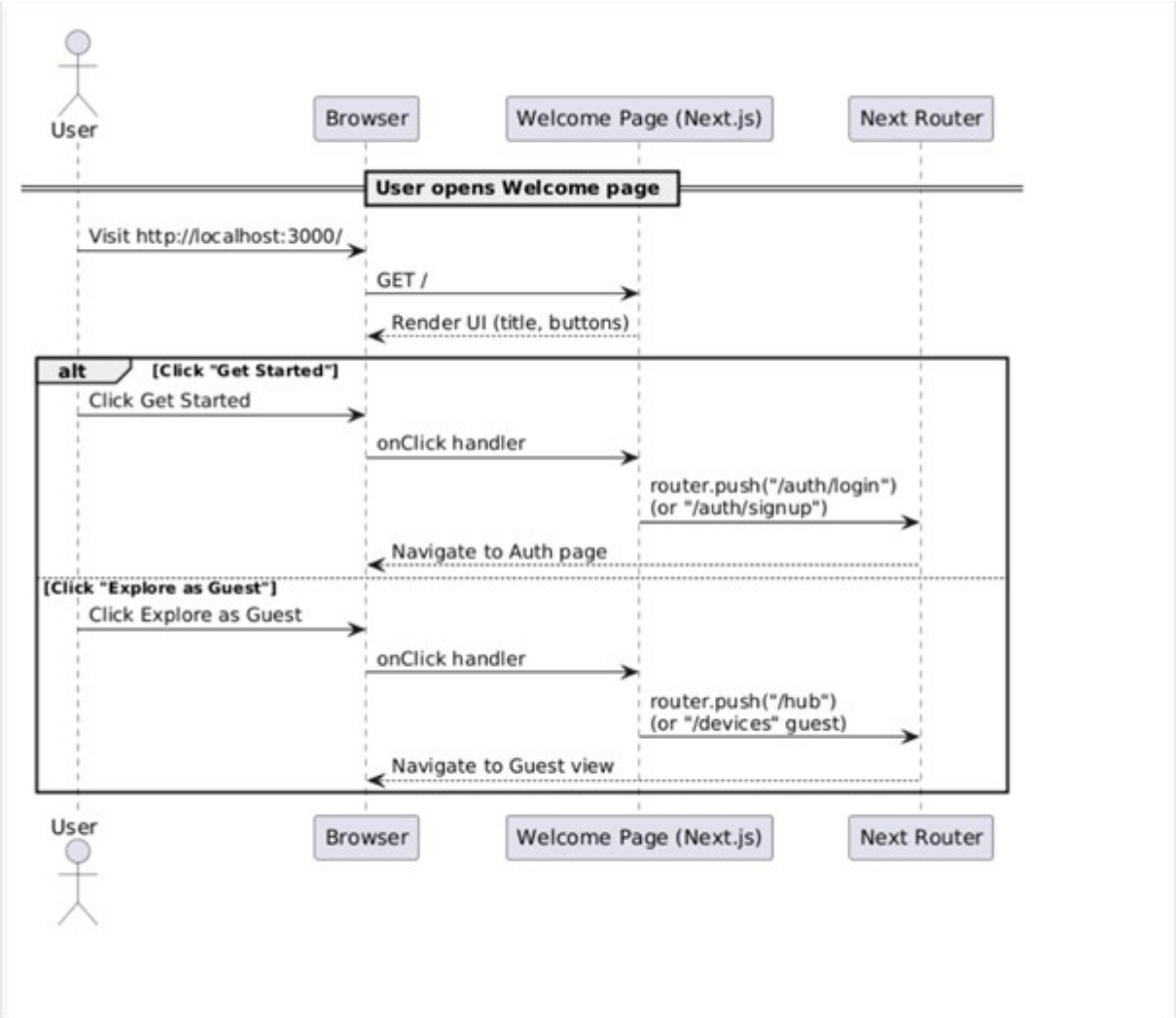
Description
First screen of the web application showing Interactive House branding and entry options.

Components
App logo + title + short tagline
Get Started button
Explore as Guest link (limited access)

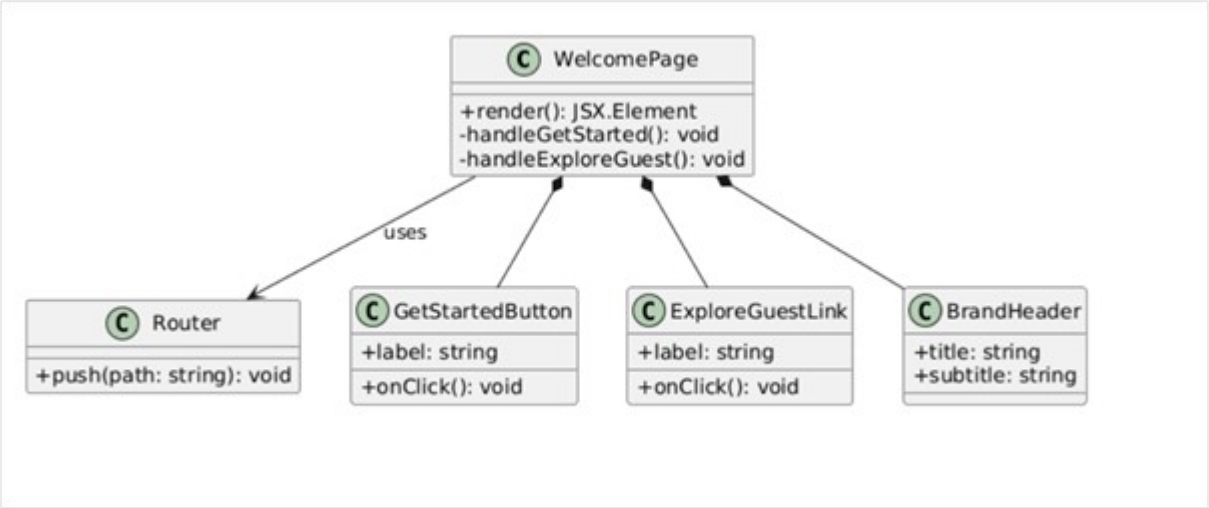
Design
Get Started routes the user to Login/Signup.
Guest mode allows viewing pages (e.g., Devices list read-only) but blocks actions like toggling actuators and database write actions.



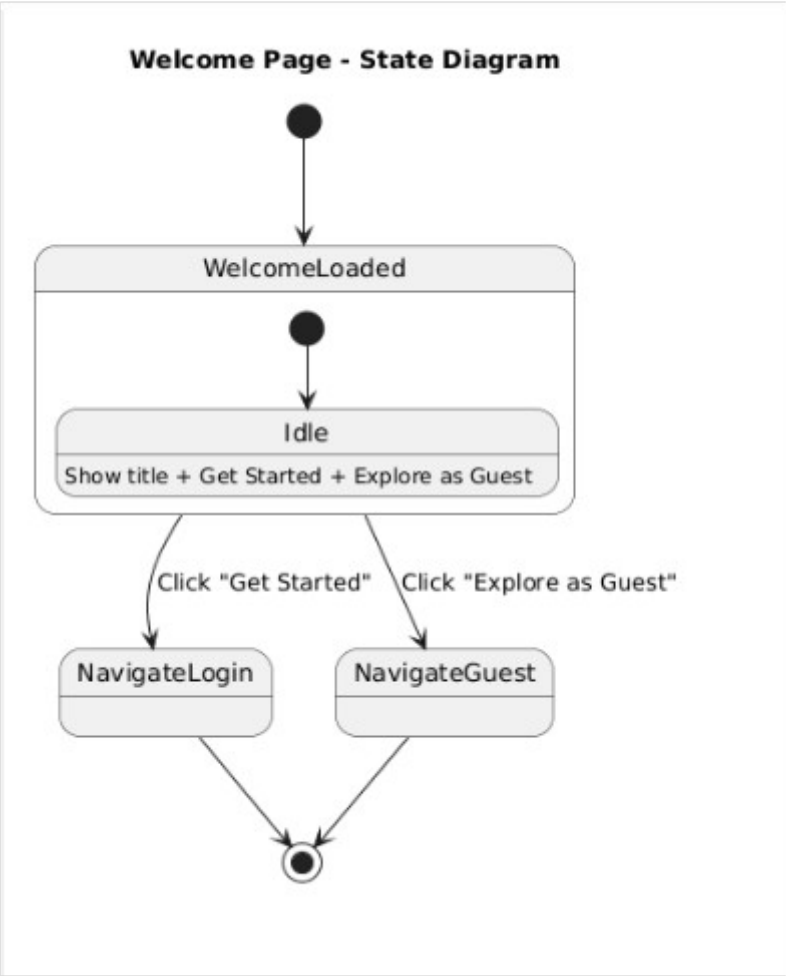
Design Item 1: Use Case Diagram, Welcome Page



Design Item 2: Sequence Diagram, Welcome Page Navigation



Design Item 3: Class Diagram, Welcome Page Structure (UI + Navigation)



Design Item 4: State Diagram, Welcome page

D2. Web User Authentication (Login/Signup)

Description

Secure login and signup for registered users to access full system features.

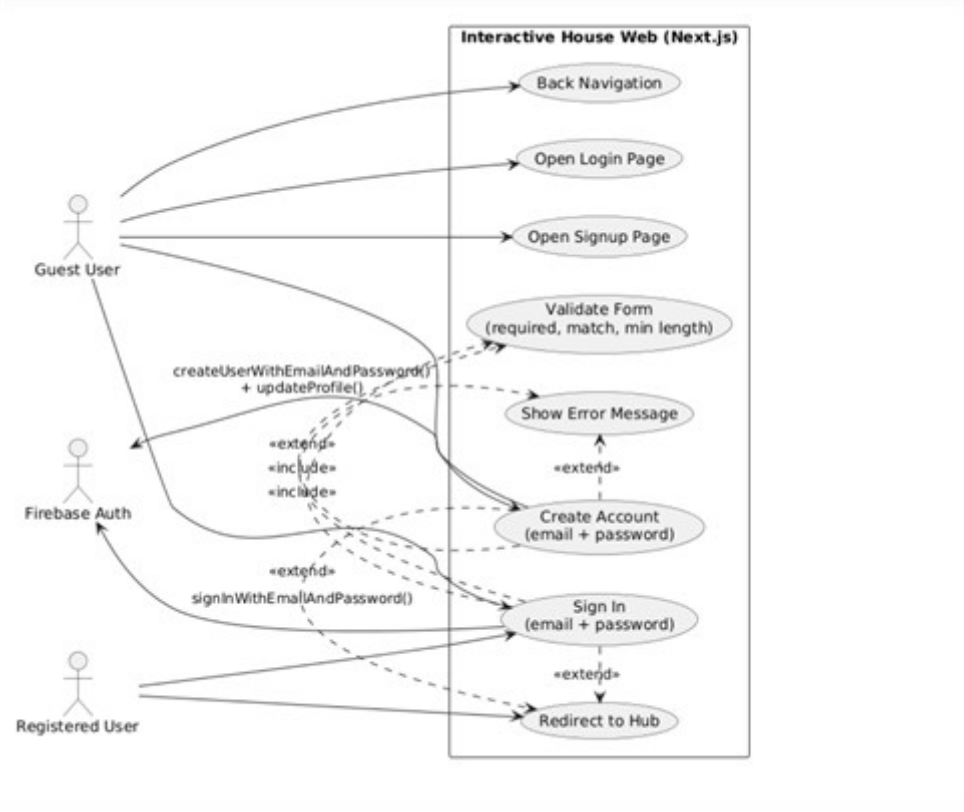
Components

Email & Password Authentication
Sign In button
Signup link

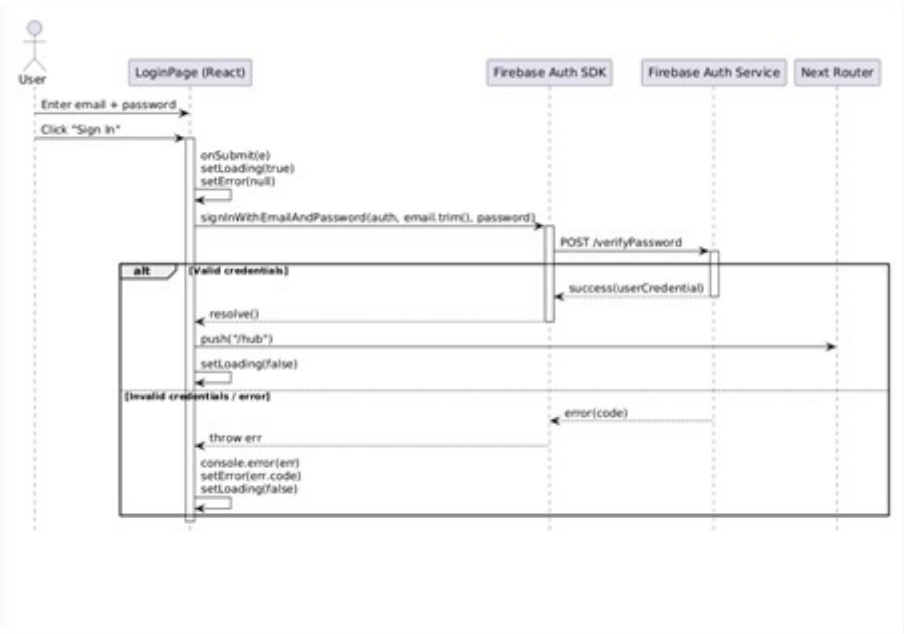
Design

Users must authenticate before performing any control actions.
Passwords are never stored in plain text; only password hashes.
After successful login: redirect to Devices or Hub screen.
Security constraints:

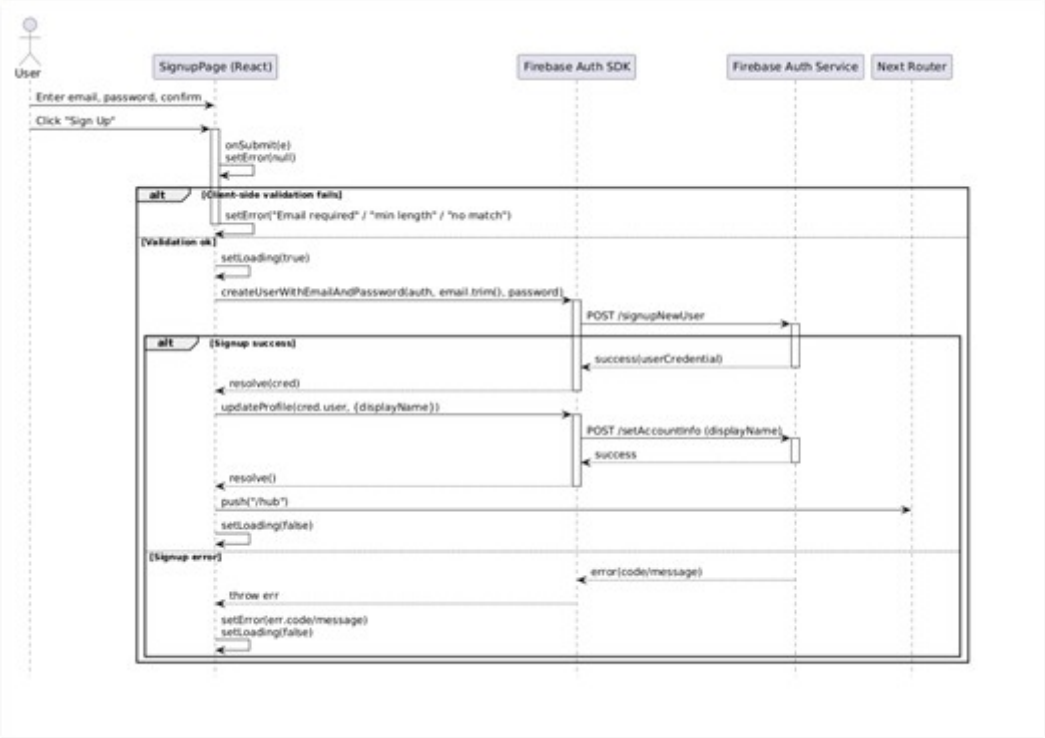
- Minimum password length (e.g., 8)
- Basic rate limiting on login attempts (server-side)



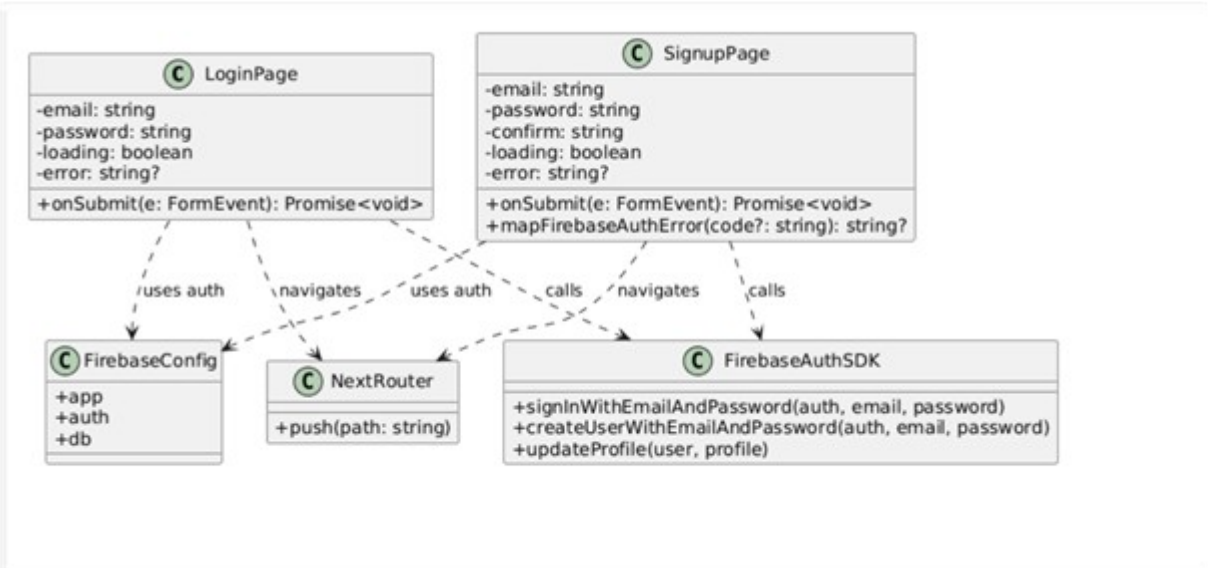
Design Item 5: Login/Signup, Use Case Diagram



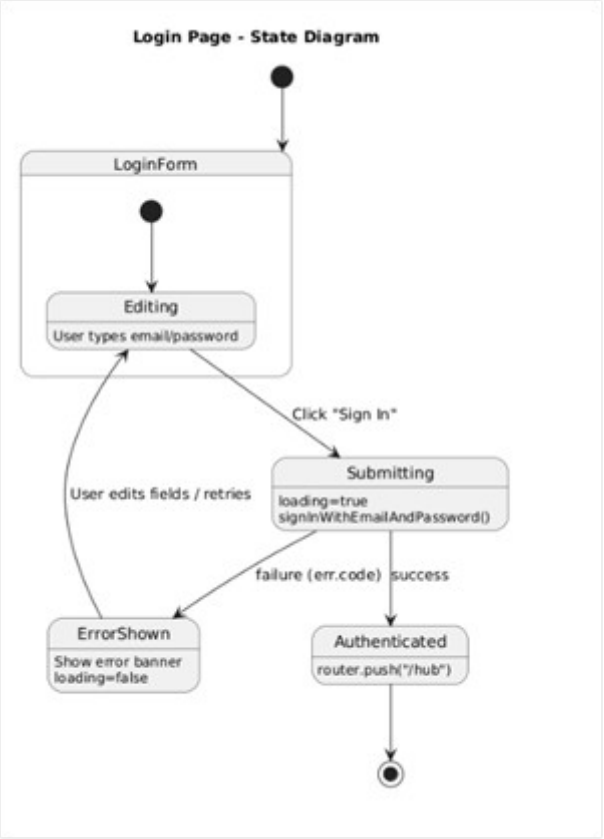
Design Item 6: Login, Sequence Diagram



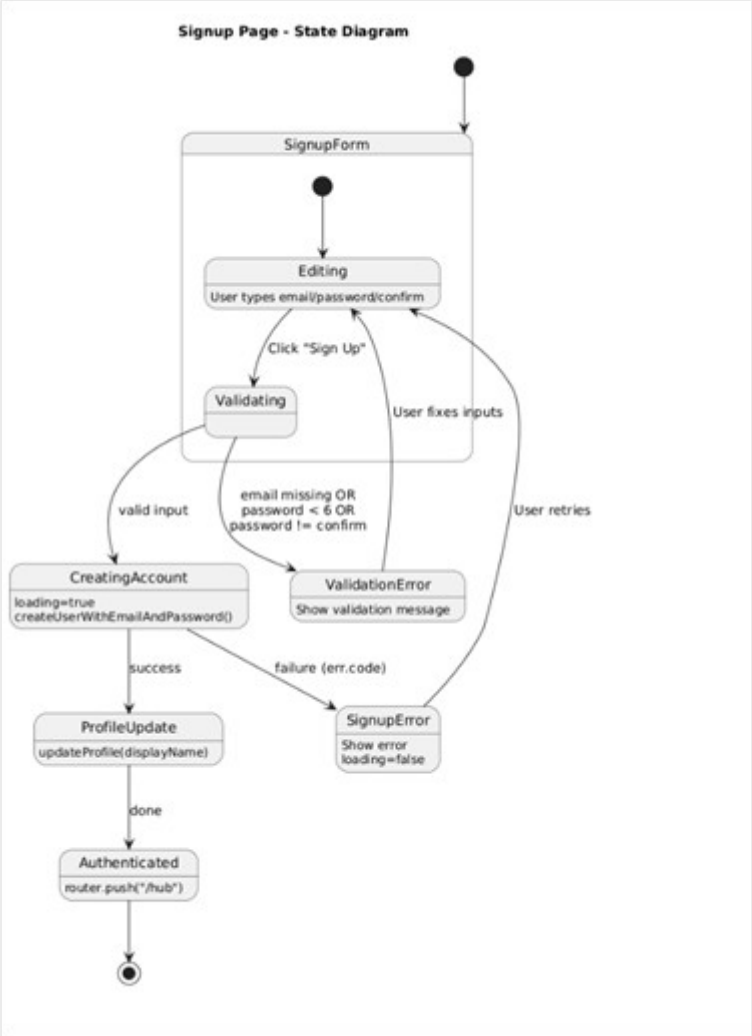
Design Item 7: Signup, Sequence Diagram



Design Item 8: Login/Signup, Class Diagram



Design Item 9: Login Page, State Diagram



Design Item 10: Signup Page, State Diagram

D3. Web Device Management Interface

Description
The Web Device Management Interface provides a centralized overview of all connected smart home devices and sensors. It allows users to view device status, hardware pin assignments, and connectivity state through a structured card based interface. The page acts as a device inventory and monitoring interface and complements the Hub control panel where actual control actions are performed.

Components

Device cards grid layout
Each device card contains:

- Device icon
- Device name (e.g., White Light, Fan INA, Door)
- Hardware pin reference (e.g., Pin 13, Pin 9)
- Current device state (ON/OFF, OPEN/CLOSED)

Navigation components:

- Top navigation header
- Links to Hub, AI, Music and Database

Design

Device data is retrieved dynamically from Firebase Firestore. The interface uses a card-based layout with dark theme styling, consistent with the mobile UI design. Each device card displays real-time information using Firebase snapshot listeners. The page functions as a monitoring interface, while device control actions are primarily executed through the Hub Control Panel. Guest mode allows users to view devices but disables any interaction with actuator controls.

D4. Web Hub Control Panel

Description

The Web Hub Control Panel serves as the main dashboard for interacting with smart home hardware devices.
It provides real time control of actuators and monitoring of sensor telemetry through a unified interface.

The hub aggregates device state information from the Firebase cloud database and allows authenticated users to update device states.

Components

Actuator control section:

- White Light toggle switch
- Orange Light toggle switch
- Fan INA control toggle
- Fan INB control toggle
- Door servo control (open/close)
- Window servo control (open/close)

Sensor monitoring section:

- Motion sensor status indicator
- Steam sensor raw value display
- Gas sensor raw value and alert indicator

System synchronization section:

- Arduino synchronization status
- Last source of update
- Last updated timestamp

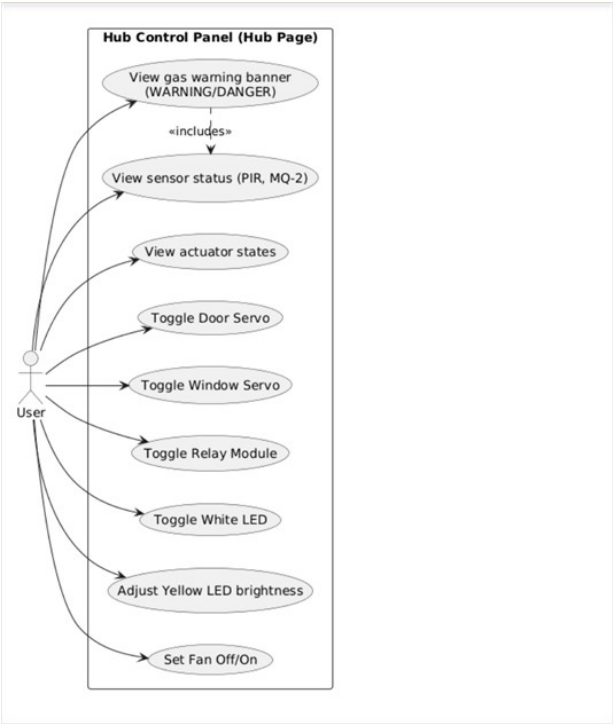
User interface elements:

- Toggle buttons for device control
- Real time sensor value display
- Visual alerts for abnormal sensor states

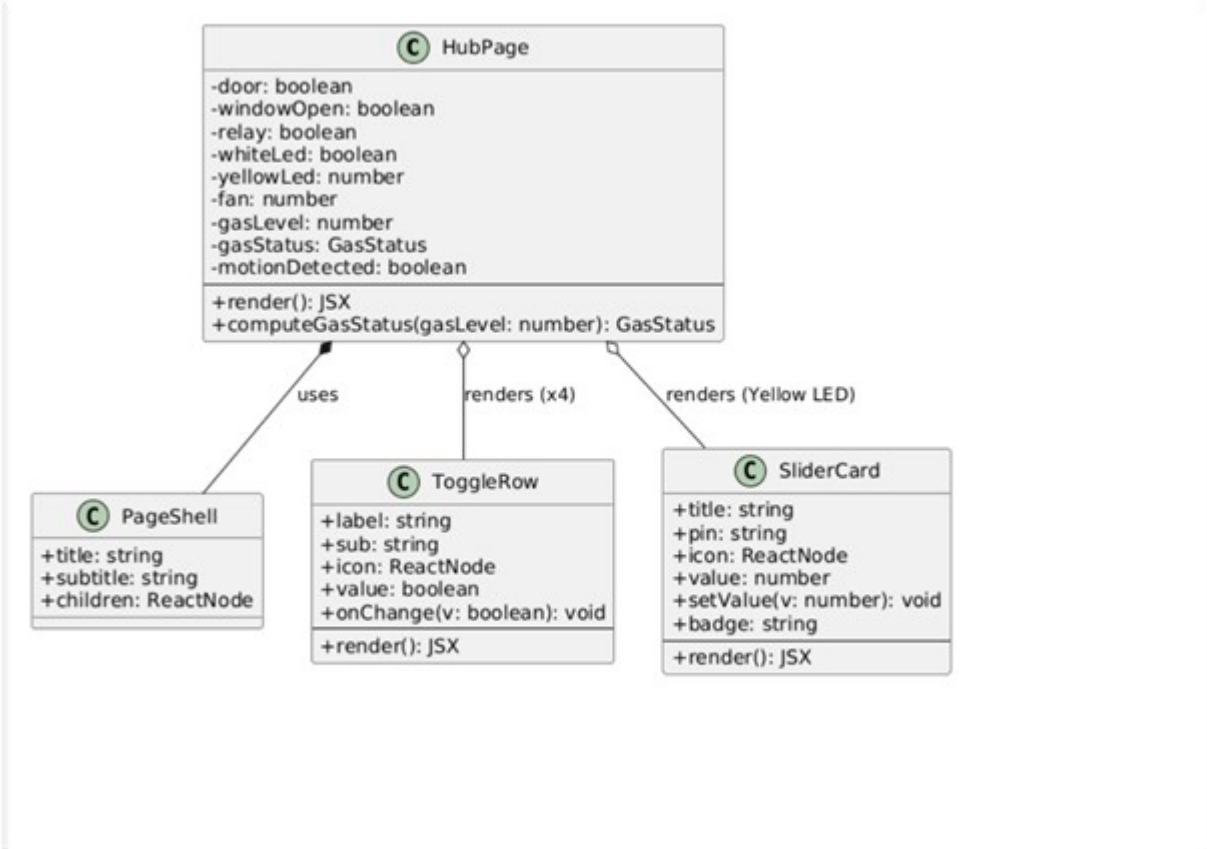
Design

The Hub Control Panel is connected to **Firebase Firestore** using real time listeners (onSnapshot).
Device states are synchronized instantly between the web dashboard and the hardware system.

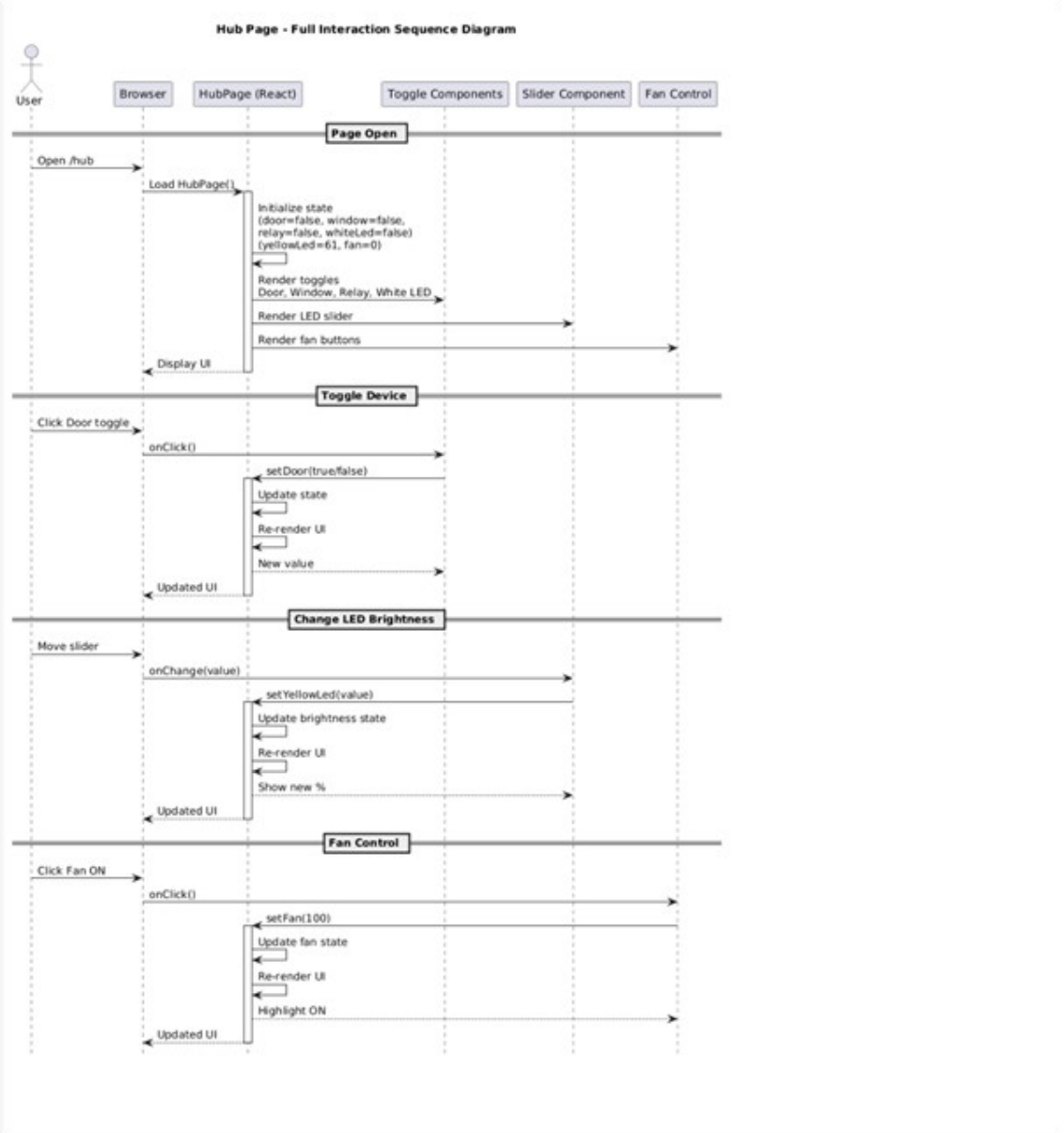
When a user interacts with a control toggle, the application sends an update request to Firestore using updateDoc(). The Arduino system reads these updates and performs the corresponding hardware actions. The UI automatically updates whenever device or sensor values change, ensuring that the dashboard always reflects the current hardware state. The design ensures real-time communication between the web interface, cloud database, and hardware layer. Access control rules are that only authenticated users can control actuators. Guest users can view sensor data but cannot modify device states.



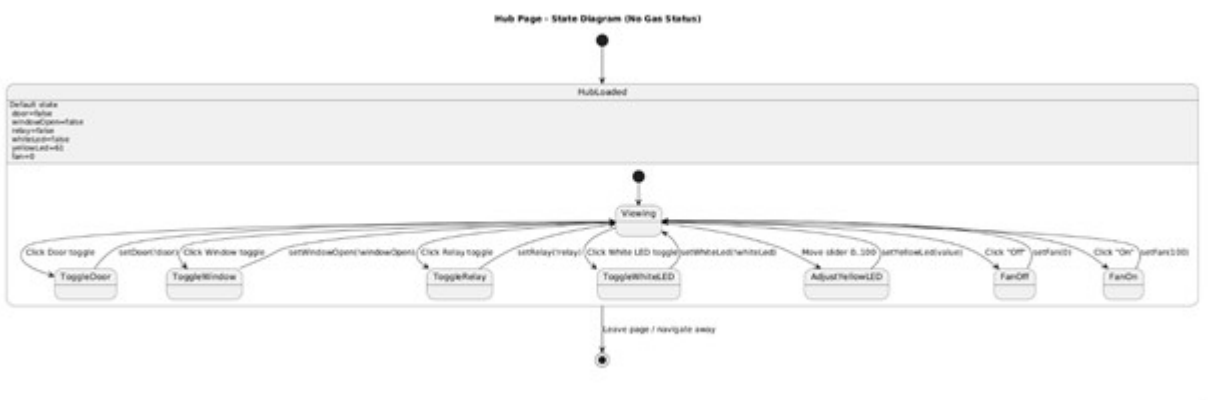
Design Item 11: Use Case Diagram, Hub Page



Design Item 12: Class Diagram, Hub Page (components + state + logic)



Design Item 13: Sequence diagram for the Hub page

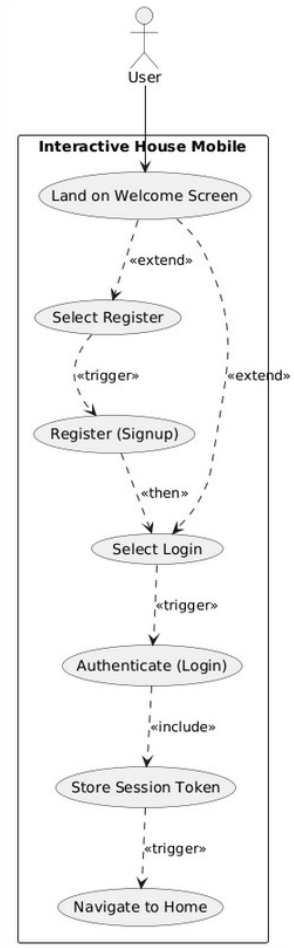


Design Item 14: Hub Page, State Diagram

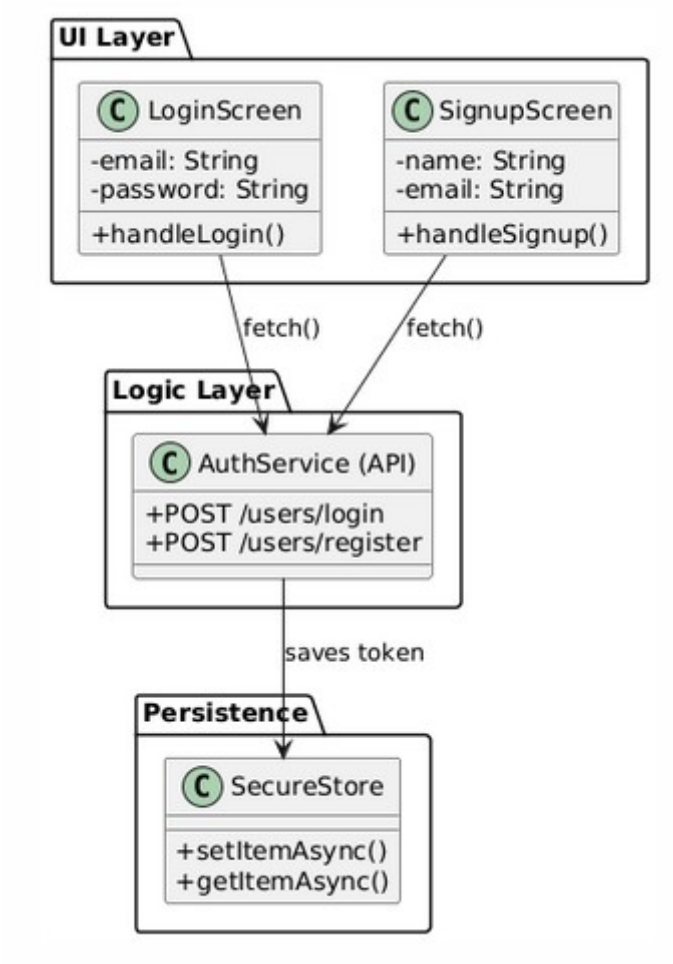
2. Mobile (Expo + React Native)

D5. Mobile Login/Signup

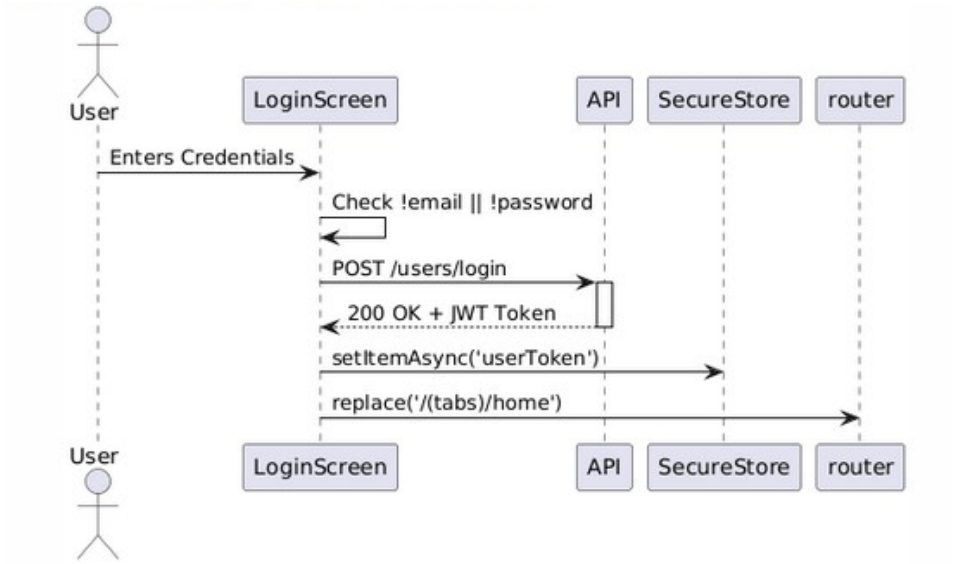
The login/signup module serves as the primary security gateway for the Interactive House Mobile application. This design item manages the transition from an unauthenticated Welcome Screen state to a fully authorized session.
The Use Case diagram illustrates the top-down functional flow, showing how the Welcome Screen extends into Signup and Login actions, ultimately triggering the Navigate to Home use case upon token storage.
The Class diagram specifies the structure of the authentication system, highlighting the data attributes (email, password, name) and core methods (handleLogin(), handleSignup()) that facilitate communication between the UI and the storage utilities.
The Sequence diagram details the chronological communication between the User, API, and SecureStore, specifically showing the logic gate where the system checks for credentials before firing a network request.
The State diagram models the application's behavior at a high level, defining the transitions between LoggedOut and LoggedIn status and the internal logic used to handle "App Start" and session timeouts.



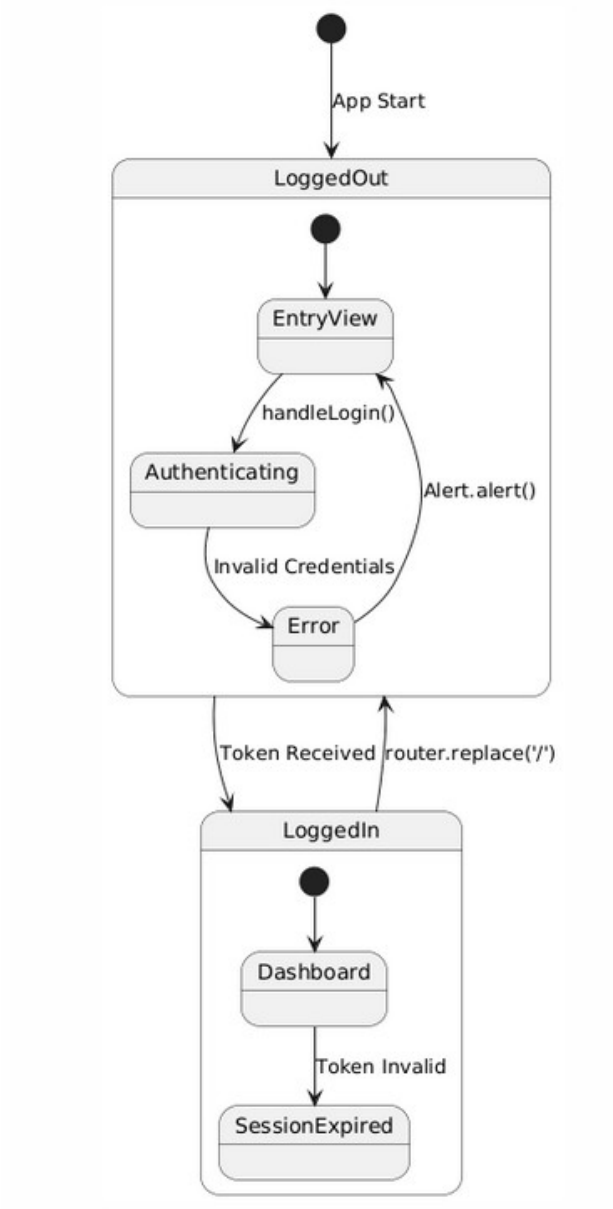
Design Item 15: Use Case Diagram



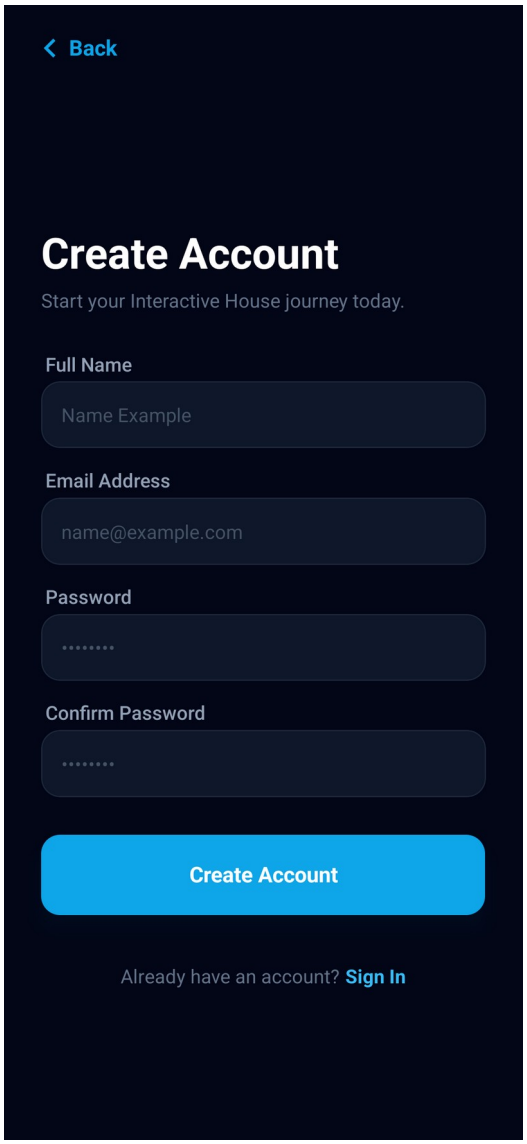
Design Item 16: Class Diagram



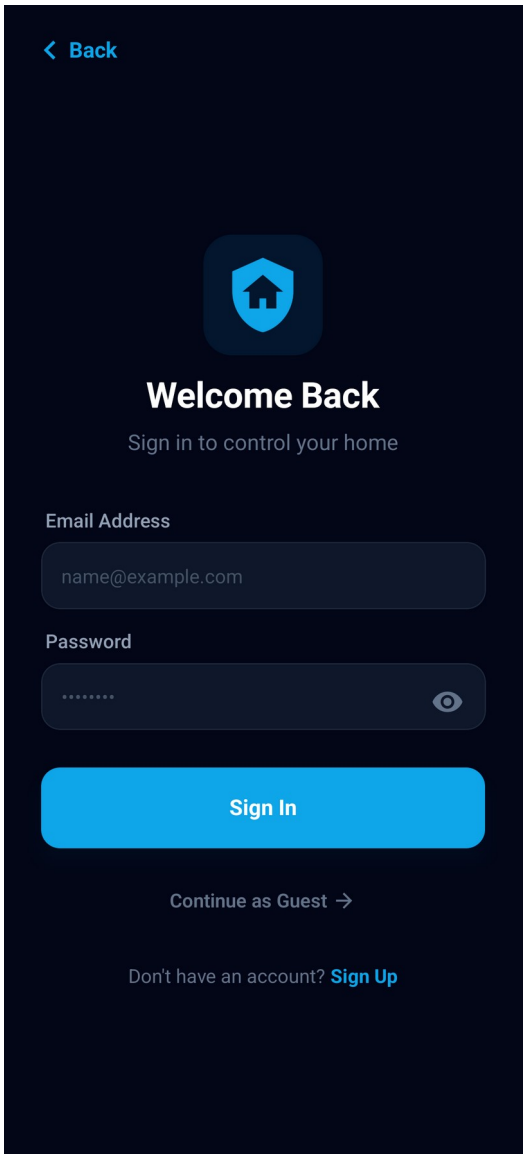
Design Item 17: Interaction Sequence Diagram



Design Item 18: State Diagram



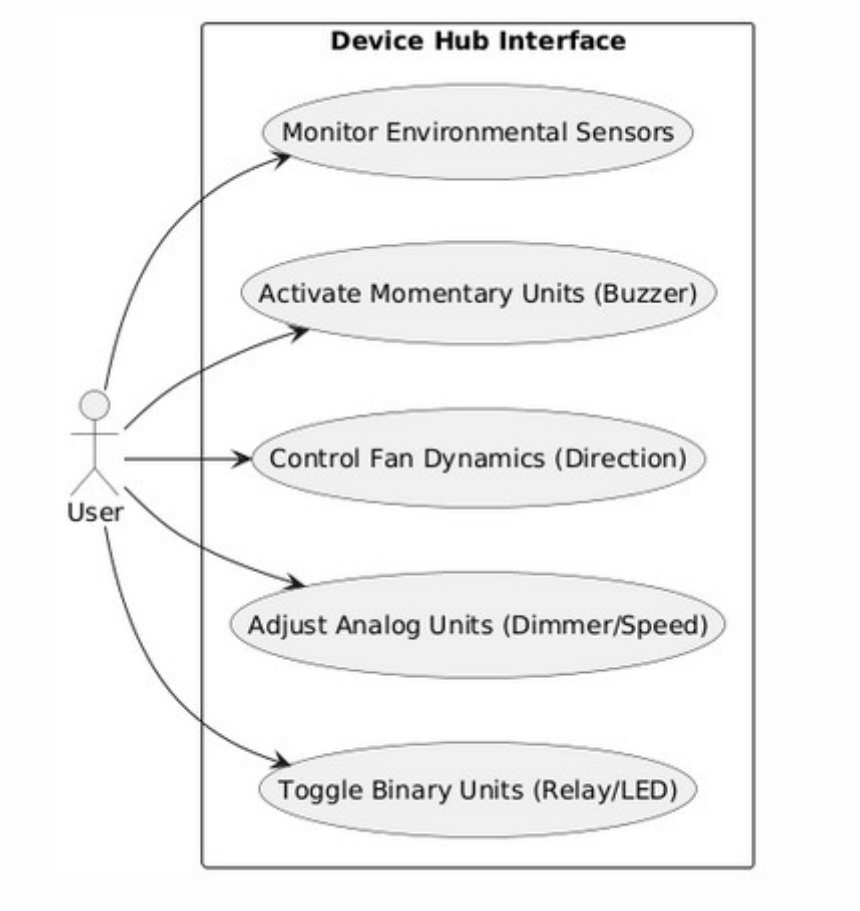
Design Item 19: Mobile App Preview – Signup Screen



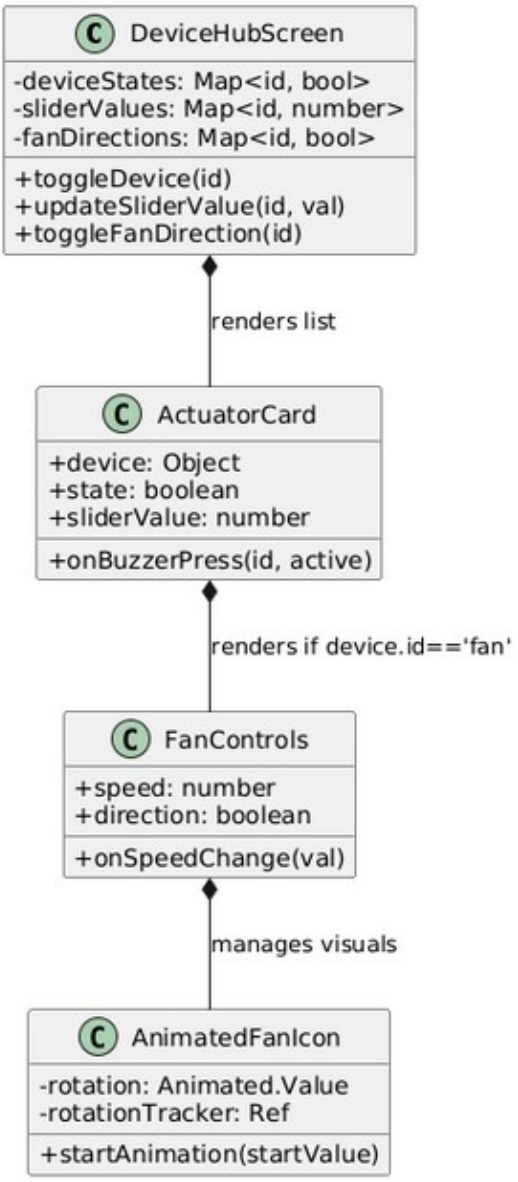
Design Item 20: Mobile App Preview – Login Screen

D6 - Mobile Control Devices in Device Hub

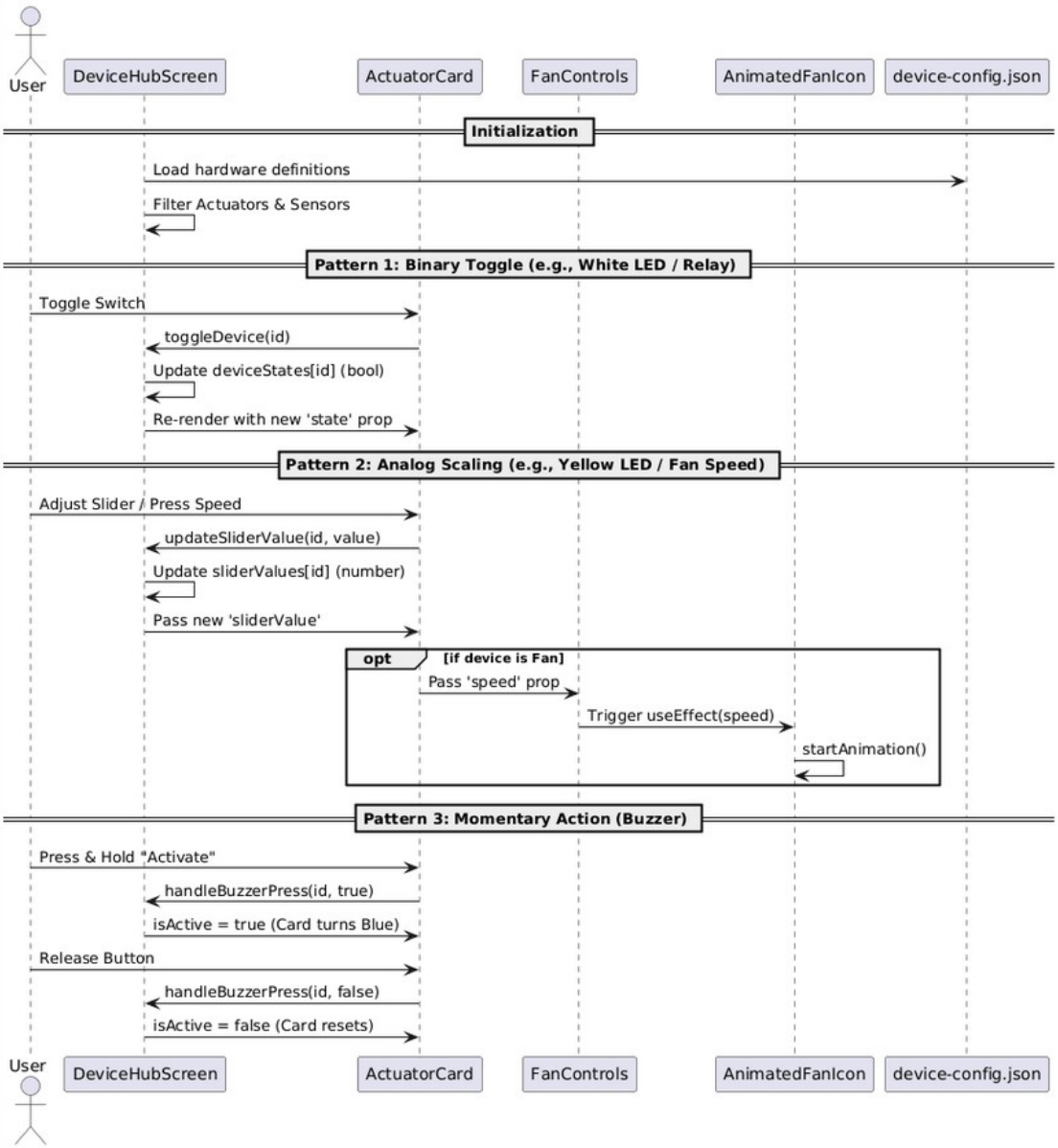
The Device Hub is the core operational component of the mobile, responsible for the real-time control and monitoring of hardware actuators and sensors. The design implements a React Native component hierarchy that translates raw hardware configuration (devices-config.json) into an interactive UI element. This will later be migrated directly through the database connection. However, Json might be good to keep, for mockup data and offline persistence.



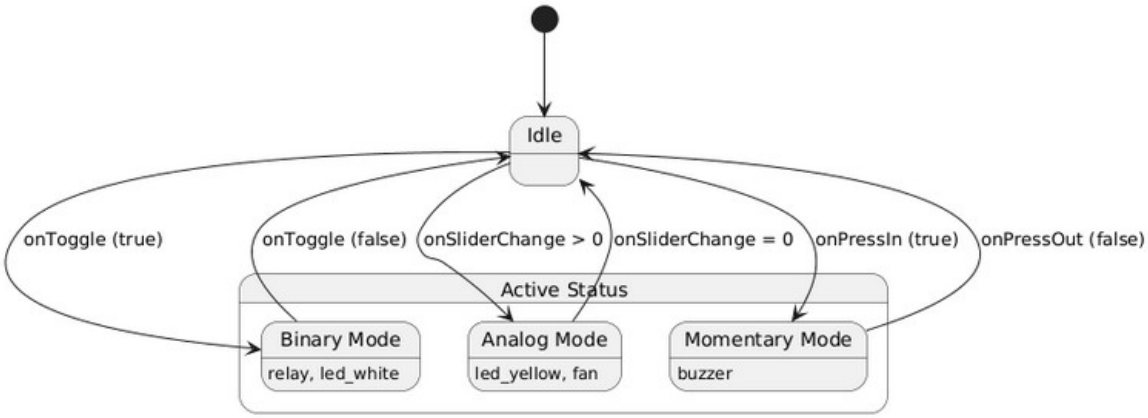
Design Item 21: Use Case Diagram



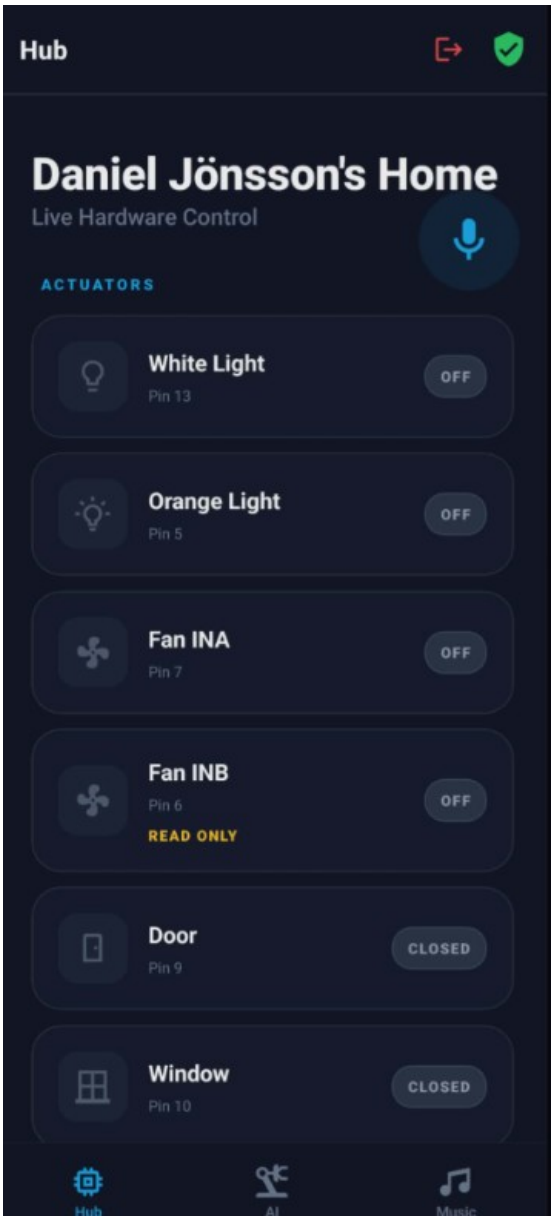
Design Item 22: Class Diagram


User
DeviceHubScreen
ActuatorCard
FanControls
AnimatedFanIcon
device-config.json
Initialization
Load hardware definitions
Filter Actuators & Sensors
Pattern 1: Binary Toggle (e.g., White LED / Relay)
Toggle Switch
toggleDevice(id)
Update deviceStates[id] (bool)
Re-render with new 'state' prop
Pattern 2: Analog Scaling (e.g., Yellow LED / Fan Speed)
Adjust Slider / Press Speed
updateSliderValue(id, value)
Update sliderValues[id] (number)
Pass new 'sliderValue'
opt
[if device is Fan]
Pass 'speed' prop
Trigger useEffect(speed)
startAnimation()
Pattern 3: Momentary Action (Buzzer)
Press & Hold "Activate"
handleBuzzerPress(id, true)
isActive = true (Card turns Blue)
Release Button
handleBuzzerPress(id, false)
isActive = false (Card resets)
User
DeviceHubScreen
ActuatorCard
FanControls
AnimatedFanIcon
device-config.json

Design Item 23: Interaction Sequence Diagram



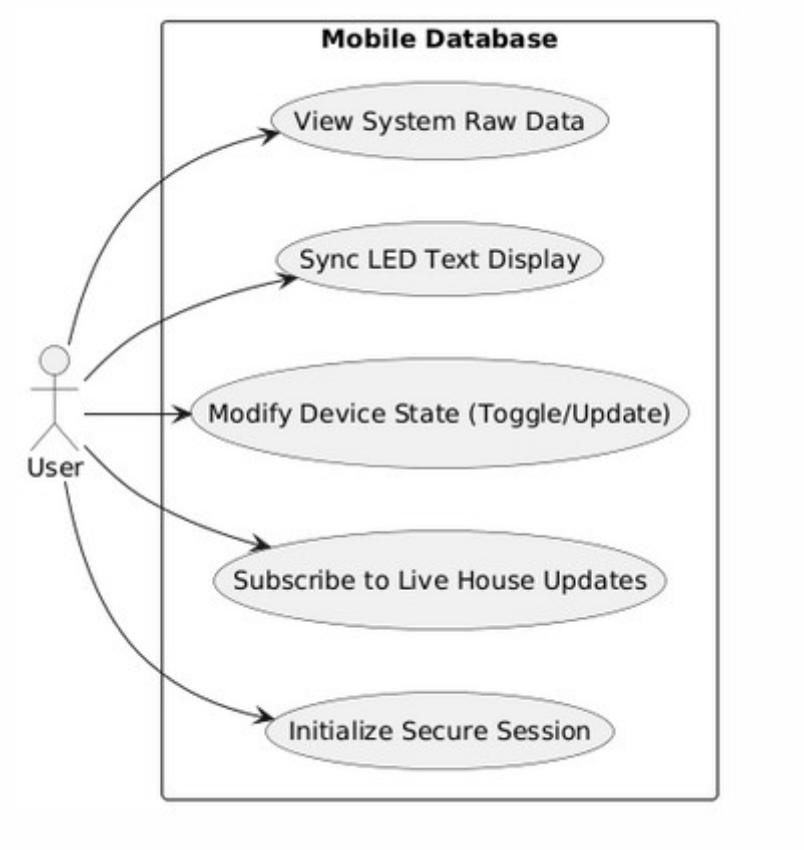
Design Item 24: State Diagram



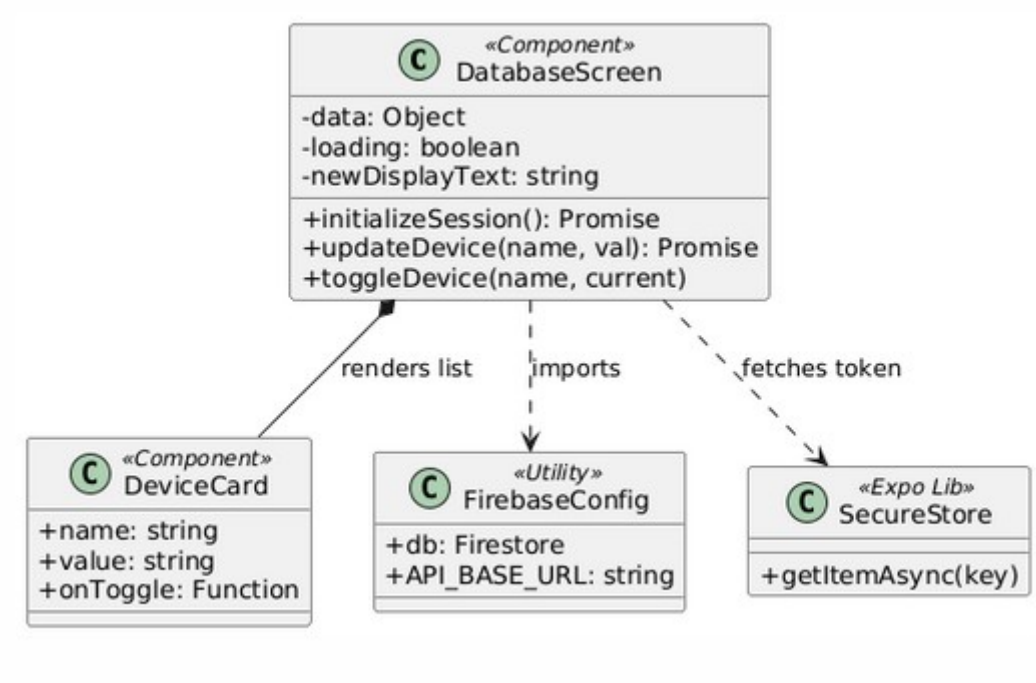
Design Item 25: Mobile App Preview of Device Hub

D7 – Mobile Database Connection

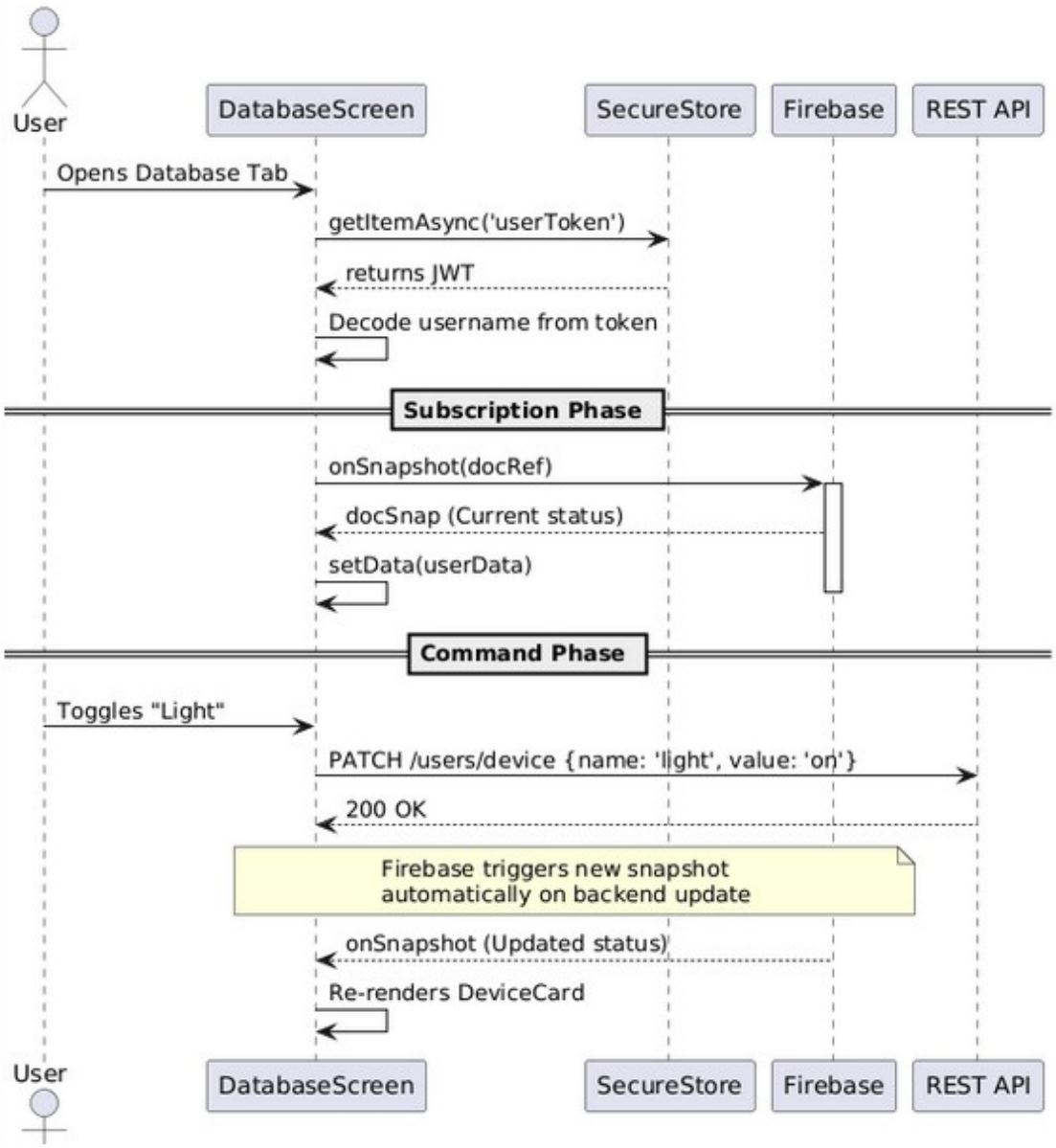
The Database Connection design item ensures data persistence and real-time state synchronization between the mobile client and the smart home hardware. This module bridges the local UI with the cloud backend through a high-performance hybrid architecture.



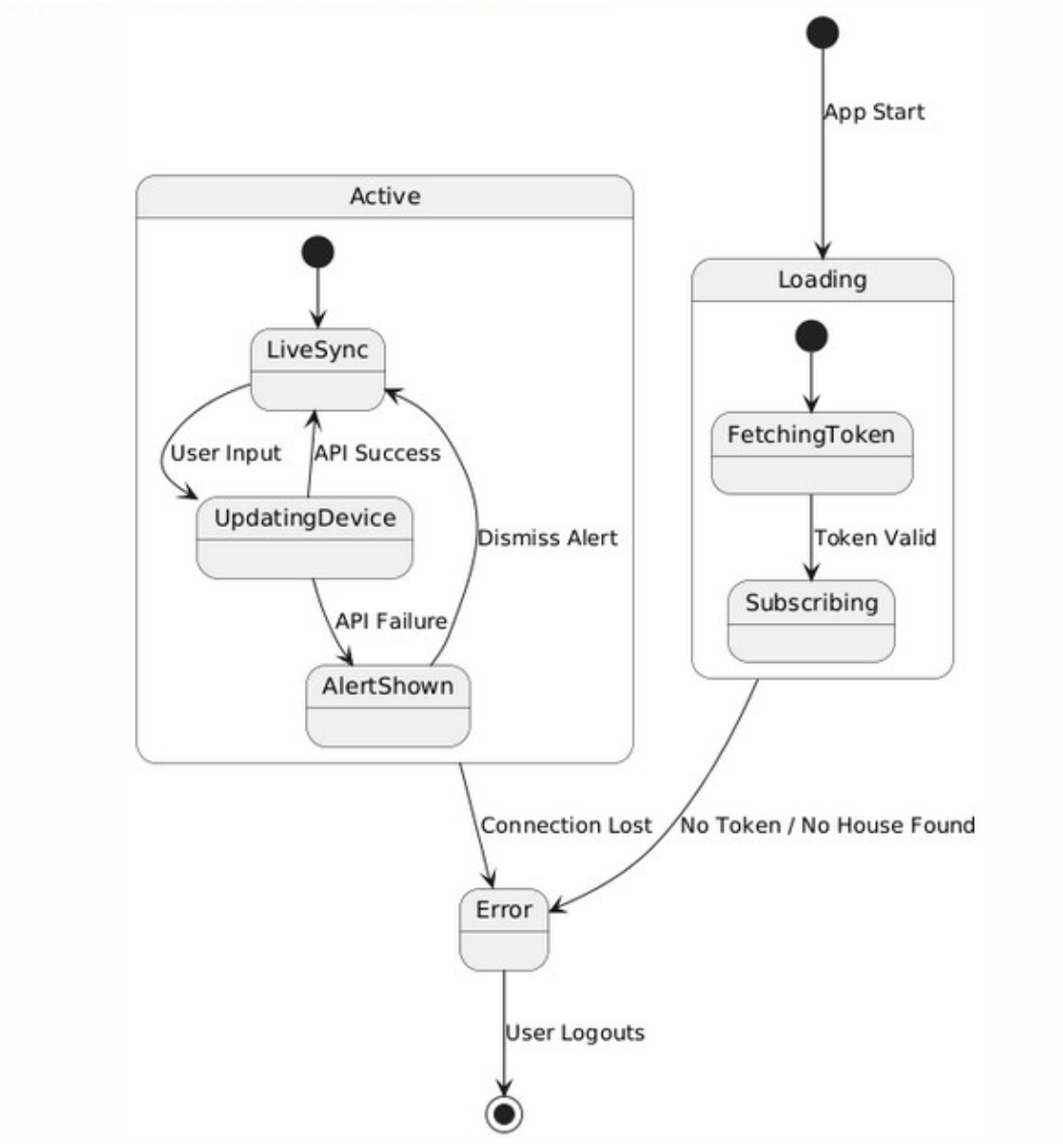
Design Item 26: Use Case Diagram



Design Item 27: Class Diagram



Design Item 28: Interaction Sequence Diagram



Design Item 29: State Diagram